Gauss-law protection and exact fourth-order lifting of a T_1^{+-} glueball band in strongly coupled $SU(N \geq 3)$ lattice gauge theory

compact localized states, a real-space sum of squares, and the all-rank band-shape theorem

Alexander Smith*

June 15, 2026

Abstract

We determine the one-excitation band structure of strongly coupled $SU(3)$ Hamiltonian lattice gauge theory in 3+1 dimensions in exact rational arithmetic, and resolve the first order at which the lowest charge-conjugation-odd band becomes mobile. At second order the lowest C-odd branch is exactly flat over the Brillouin zone. Its flat subspace is $\ker B(k)^\dagger$, where the signed hopping symbol factors as $\widetilde{N}(k) + 4I = B(k)B(k)^\dagger$; compact localized states are the consistently oriented boundaries of elementary cubes, and flatness is the chain-complex identity $\partial_2\partial_3 = 0$. All third-order tromino processes vanish by a bare-link lemma, so the band remains rigid through $O(y^3)$:

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 + O(y^4).$$

Using the complete Hermitian 189-record fourth-order real-space kernel, we show that this protection fails at the next order in a fully controlled way. For $X_i = 1 - \cos k_i$ and the cell-gauge flat vector $\psi = (e^{ik_2} - 1, -(e^{ik_1} - 1), e^{ik_0} - 1)^T$, the projected coefficient factorizes exactly as

$$\psi^\dagger[H_4(k) - qI]\psi = \frac{5}{12}\sum_i X_i^2 + \frac{17607806155349}{275331901291200}\sum_{i<j} X_iX_j,$$

where $\|\psi\|^2 = 2\sum_i X_i$ and

$$q = -\frac{20721577909065127111}{7250590288602460800}.$$

This momentum-space factorization lifts exactly to a local real-space sum-of-squares theorem. If $C = \partial_3$ is the oriented cube-boundary map, T_i translates cube amplitudes, and $L_i = (T_i - I)^\dagger(T_i - I)$, then

$$C^\dagger(H_4 - qI)C = \frac{5}{48}\sum_i L_i^2 + \frac{17607806155349}{1101327605164800}\sum_{i<j}(\nabla_i\nabla_j)^\dagger(\nabla_i\nabla_j).$$

Thus the fourth-order mobility is carried by axial three-cube second differences and planar four-cube checkerboard differences. Positivity proves that Γ is the unique global minimum; elementary cube inequalities prove that $R = (\pi, \pi, \pi)$ is the unique global maximum. The

*Independent Researcher. Strong-coupling variable $y = \beta/6 = 1/g_H^4$ throughout. Every displayed strong-coupling constant, exact enumeration count, and string-tension coefficient is machine-certified in exact (rational or finite-field) arithmetic; the verification chain, certificate hashes, and reproduction steps are collected in Appendix A, and the determinant-sector σ_5 certificate in Appendix B.

fourth-order bandwidth is $\Delta W_4 \approx 0.4806178691$, with its exact rational value given below. The lift near Γ is quadratic but direction-dependent, so no ordinary effective-mass Hessian exists there; near R the curvature is isotropic and negative. Thus the Gauss-law band is exactly immobile through third order and acquires its first nonzero, parameter-free dispersion at fourth order. Independently, the shared-link calculus extends to every $SU(N)$ with integer $N \geq 3$: in the reduced normalization fixed by the $SU(3)$ domino,

$$t_N = \frac{2N(N^2-4)}{(N^2-1)(4N^2-9)(2N^2-1)} > 0, \qquad t_N \sim \frac{1}{4N^3},$$

and the group-independent incidence kernel gives a universally flat closed-surface branch at second order. At fourth order the complete $SU(3)$ source chain is recovered from Stage 0 through the folded des-Cloizeaux Stage 3I and final kernel assembly Stage 3J. For every integer $N \geq 7$, exact walled-Brauer contraction of the 4,171 ordered words gives

$$D_N(k) = A_N \sum_i X_i^2 + B_N \sum_{i<j} X_i X_j, \quad q_N = -\frac{2}{3N}\frac{Q_{32}(N^2)}{D_{34}(N^2)},$$

$$A_N = \frac{640}{N(N^2-1)^3}, \qquad B_N = \frac{P_{17}(N^2)}{NR_{20}(N^2)}.$$

Exact denominator factorizations and positive Newton expansions prove $q_N < 0$, $A_N > 0$, and $B_N > 0$ in the stable range. A separate exact N-ality scan and determinant-sector contraction then closes the band shape at the exceptional ranks: the $SU(4)$ determinant contribution satisfies $\Delta A_4 = \Delta B_4 = 0$, no determinant assignment occurs for $SU(5)$, and the sole $SU(6)$ determinant orbit is absent from the A/B target. Together with the rank-specific $SU(3)$ coefficients, this proves a unique minimum at Γ, a unique maximum at R, and bandwidth $A_N + B_N > 0$ for every integer $N \geq 3$. The determinant offsets q_4, q_6 are given. In pure $SU(2)$ charge conjugation is a gauge transformation, so the C-odd sector is empty and $N \geq 3$ is the maximal domain. Continuing to fifth order in $SU(3)$, the projected coefficient keeps the identical two-invariant form with $A_5 = 313/240$ and $B_5 > 0$, so the unique-minimum/unique-maximum structure persists through $O(y^5)$. Combined with the project-native Hamiltonian string tension (bridge $\sigma(y) = \frac{1}{2}W(2y)$) this gives the scale-matched ratio $m_{1^{+-}}/\sqrt{\sigma}$ through $O(y^5)$. The strong-coupling series alone does not reach the continuum—its fourth-order truncation peaks well below the physical ratio and standard resummations fail—but the exact $1/N^2$ band structure supplies the functional form for a held-out extrapolation of continuum lattice data across rank, reproducing the independently measured large-N value and predicting $m_{1^{+-}}/\sqrt{\sigma}|_{SU(3)} \approx 6.15$ against the measured 6.065(40). The C-even band and the T_1^{+-} quantum-number assignment are also given exactly. No continuum mass is claimed from the series itself; the large-N estimate is a lattice-data extrapolation made trustworthy by the exact $1/N^2$ form.

1 Introduction

Flat bands — single-particle dispersions that are exactly constant over the Brillouin zone — arise from destructive interference encoded in lattice geometry and are spanned by compact localized states (CLS) [7, 8, 9, 10, 11, 12]. They are usually discussed in engineered condensed-matter, photonic, and cold-atom systems. Here the same structure appears without geometric engineering in strongly coupled $SU(3)$ Hamiltonian lattice gauge theory [1].

In the strong-coupling regime the elementary one-flux excitations are single plaquettes of chromoelectric flux. Degenerate perturbation theory turns the three plaquette orientations into a 3×3 Bloch problem. The main results are:

(A) At second order the lowest C-odd band is exactly flat, with coefficient $11/306$ at every momentum (Theorem 5.3).

(B) The flat subspace is a lattice Gauss-law kernel. The signed adjacency factors through the plaquette-boundary incidence map, and the compact localized states are oriented boundaries of elementary cubes (Theorems 5.1 and 5.4).

(C) Every third-order tromino amplitude vanishes by a bare-link argument. The flat band therefore shifts rigidly by the exact coefficient $-109151/249696$ (Theorem 6.2).

(D) The complete fourth-order kernel breaks the flatness. Its projection onto the lower-order flat branch factorizes into two positive cubic invariants in $X_i = 1 - \cos k_i$ (Theorem 7.3).

(E) The projected operator has an exact local real-space sum-of-squares representation. Its two elementary deformations are an axial three-cube second difference and a planar four-cube checkerboard difference (Theorem 7.4).

(F) The factorization gives an analytic global theorem: Γ is the unique band minimum, R the unique band maximum, and the exact fourth-order bandwidth is $\Delta W_4 = A + B$ (Theorem 7.6).

(G) The rest-point lift is direction-dependent. The Γ band coefficient is continuous and differentiable but has no ordinary Hessian; the R curvature is isotropic and negative (Proposition 7.8).

(H) The rest-frame C-odd triplet transforms as T_1^{+-}, while the C-even sector splits as $A_1^{++} \oplus E^{++}$ and disperses normally (Theorems 5.3 and 5.5).

(I) The signed shared-link hopping is positive for every $SU(N)$, $N \geq 3$, with exact rational coefficient t_N and large-N behavior $t_N \sim 1/(4N^3)$. Since the incidence kernel is group-independent, the closed-surface second-order branch is universally flat (Theorem 5.2).

(J) At fourth order, the full stable-rank contraction is closed for every integer $N \geq 7$. Exact walled-Brauer trace wiring gives q_N, $A_N = 640/[N(N^2-1)^3]$, and the canonical compact form $B_N = P_{17}(N^2)/[NR_{20}(N^2)]$; exact Newton-positivity certificates prove $q_N < 0$, $A_N > 0$, and $B_N > 0$ (Theorem 7.11).

(K) Exact determinant-sector reductions extend the band-shape result to every integer $N \geq 3$: $SU(4)$ contributes zero to both invariants, $SU(5)$ has no determinant assignments, and the sole $SU(6)$ determinant orbit misses the A/B target. Hence Γ and R are the unique global edges and $A_N + B_N > 0$ for all non-pseudoreal ranks (Theorem 7.13).

The physical statement is therefore finite-order and sharp. A closed-surface T_1^{+-} excitation cannot propagate through the link-mediated channels that occur through $O(y^3)$: every boundary-link amplitude cancels exactly. At $O(y^4)$, site-mediated geometries activate and the cancellation no longer protects the projected branch. The resulting mobility is not inferred from a scan; it is fixed by an exact positive rational factorization.

The second-order calculation also corrects the C-even nearest-neighbor hop quoted in the companion manuscript [14]. The missing vacuum-mediated route changes it to $t_+ = -481/612 + 3/4 = -11/306$, independently adjudicated by exact two-plaquette diagonalization. The corresponding per-bond ratio $|t_-|/|t_+| = 5/22$ does not itself explain immobility; the incidence kernel does.

Strong-coupling expansions of Hamiltonian lattice gauge theory have a long history [1, 2, 3, 4], and the continuum glueball spectrum is accurately known from Euclidean Monte Carlo [5, 6]. The present statements are exact order-by-order results in the strong-coupling Hamiltonian expansion. They do not predict a continuum mass, do not establish convergence of the series, and do not claim all-orders flatness.

Section 2 fixes conventions. Sections 3 and 4 derive the one-plaquette and shared-link constants. Section 5 proves the second-order band structure and compact localized states. Section 6 proves third-order rigidity. Section 7 gives the link-channel robustness theorem, solves the $SU(3)$ fourth-order band analytically, derives its exact real-space SOS operator, and proves the complete stable-rank $SU(N \geq 7)$ fourth-order band theorem. Section 9 summarizes the exact data and physical consequences; Section 11 lists the remaining problems.

2 Setting

We work with the Kogut–Susskind Hamiltonian [1] for $SU(3)$ on a cubic spatial lattice ($d = 3$),

$$H_\beta \;=\; \frac{1}{2}\sum_\ell C_2(\ell) \;+\; \beta \sum_p \Big(1 - \tfrac{1}{N}\,\mathrm{Re}\,\mathrm{Tr}\,U_p\Big), \qquad N = 3, \tag{1}$$

in units of the electric coupling; $C_2(\ell)$ is the quadratic Casimir on link ℓ and U_p the oriented plaquette holonomy. We use the natural strong-coupling variable

$$y \;=\; \tfrac{\beta}{6} = \tfrac{1}{g_H^4}, \qquad \beta = 6y, \tag{2}$$

in which the one-plaquette energy scale is 8/3 and the perturbation per plaquette carries one power of y. The strong-coupling vacuum is the electric ground state $|0\rangle$ (all links in the trivial representation); the elementary excitations are single-plaquette flux states. With $\chi_p = \mathrm{Tr}\,U_p$ the fundamental character on plaquette p, the normalized one-flux states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}}\,|0\rangle, \qquad C|p, \pm\rangle = \pm|p, \pm\rangle, \tag{3}$$

with unperturbed energy 8/3 (four links in the fundamental representation: $4 \times \frac{1}{2} \cdot \frac{4}{3} = \frac{8}{3}$; we keep the conventions of [14] throughout). On the cubic lattice plaquettes carry three orientations $(\mu\nu) \in \{xy, yz, zx\}$, so each momentum sector is a 3×3 problem.

Degenerate strong-coupling perturbation theory in the one-flux manifold produces, order by order in y, an effective hopping Hamiltonian $H_{\mathrm{eff}}(k)$ per charge sector. Two structural facts organize the computation: (i) within-plaquette processes renormalize the diagonal and are captured exactly by the one-plaquette towers of Section 3; (ii) the leading inter-plaquette processes are mediated by *shared links* — each plaquette has exactly $4(2d - 3) = 12$ shared-link neighbors in $d = 3$ (four coplanar, eight transverse) — and reduce to balanced Haar moments on the shared link [14]. We use the linked-cluster bookkeeping of [14] (its Assumption 6.5); all finite-system counting is verified non-perturbatively by exact diagonalization on the two-plaquette "domino" (Section 4).

3 One-plaquette towers and the bridge

The $SU(3)$ one-plaquette class Hamiltonian $h_\beta = \frac{1}{2}C_2 + \beta(1 - \frac{1}{3}\mathrm{Re}\,\mathrm{Tr}\,g)$ restricted to class functions has charge-even and charge-odd gaps $\Delta_\pm(\beta)$ with exact strong-coupling towers [14]

$$\Delta_\pm(\beta) = \tfrac{2}{3} \mp \tfrac{\beta}{6} + b_2^\pm \beta^2 + b_3^\pm \beta^3 + O(\beta^4), \qquad b_2^+ = \tfrac{13}{180},\ b_3^+ = \tfrac{101}{2700},\ b_2^- = \tfrac{1}{18},\ b_3^- = \tfrac{7}{432}. \tag{4}$$

The bridge to the lattice one-flux sector evaluates the one-plaquette gap at the local class coupling $b = \beta/4 = 3y/2$ (one quarter of the lattice coupling), with an overall factor 4: the *within-plaquette* part of the effective one-flux energy is $4\Delta_\pm(3y/2) = 4\Delta_\pm(\beta/4)$, i.e.

$$\underbrace{\tfrac{8}{3} \mp y + \tfrac{13}{20}y^2 + \tfrac{101}{200}y^3}_{C\text{-even}}, \qquad \underbrace{\tfrac{8}{3} + y + \tfrac{1}{2}y^2 + \tfrac{7}{32}y^3}_{C\text{-odd}}, \tag{5}$$

to the orders needed here. Everything beyond (5) is inter-plaquette leakage, computed in the next two sections.

4 Second order: exact leakage and the corrected C-even hop

At $O(y^2)$ the only inter-plaquette geometry is the shared-link pair. Integrating the six free links of the pair by the fusion identity $\int dB\ \mathrm{Tr}(XB)\,\mathrm{Tr}(YB^\dagger) = \mathrm{Tr}(XY^\dagger)/3$ reduces every required quantity to balanced $(2,2)$ Haar moments on the shared link, resolved into the irreducible channels $\{1, 8, \bar{3}, 6\}$ of the fused link representation, with dimension-weighted norms $N_R = d_R/9$. The corresponding intermediate energies are $\{4, \frac{11}{2}, \frac{14}{3}, \frac{17}{3}\}$, to be used in the denominators $E_0 - E_R$ with $E_0 = 8/3$ [14].

Lemma 4.1 (Channel sums)**.** *Per shared-link neighbor, the fused-channel second-order sum is*

$$\Sigma_{\text{chan}} = -\frac{1}{12} - \frac{16}{51} - \frac{1}{6} - \frac{2}{9} = -\frac{481}{612}, \tag{6}$$

independent of the relative link orientation, and equal for the self-energy and hopping networks (the operator identity $\langle G_{p'}|W_p = \langle G_p|W_{p'}$ of [14]). The diagonal carries in addition the vacuum subtraction $+3/4$ per neighbor.

The companion manuscript identified Σ_{chan} with the full hop. It is not: the hop has one more intermediate state.

Lemma 4.2 (Vacuum-mediated route)**.** *For C-even one-flux states the second-order effective Hamiltonian contains the vacuum-intermediate route*

$$\langle e_r|W\,RW|e_i\rangle \supset \frac{\langle e_r|W|0\rangle\langle 0|W|e_i\rangle}{8/3 - 0} = (\sqrt{2})(\sqrt{2})\,\frac{3}{8} = +\frac{3}{4} \tag{7}$$

for every *ordered pair (i, r), adjacent or distant. It is absent in the C-odd sector ($\langle o|W|0\rangle = 0$ by parity). For distant pairs it exactly cancels the free two-flux channel sum $2/(8/3 - 16/3) = -3/4$, recovering the vanishing of distant hops; for shared-link neighbors the cancellation against Lemma 4.1 is partial.*

Theorem 4.3 (Second-order leakage, both sectors)**.** *Per shared-link neighbor:*

$$C\text{-even:}\quad \textit{self-energy} = -\tfrac{481}{612}, \quad \textit{vacuum subtraction} = +\tfrac{3}{4}, \quad t_+ = -\tfrac{481}{612} + \tfrac{3}{4} = -\tfrac{11}{306}; \tag{8}$$

$$C\text{-odd:}\quad \textit{self-energy} = -\tfrac{481}{612}\ (C\textit{-independent}), \quad t_-(s) = \tfrac{5s}{612}, \tag{9}$$

where $s = \pm 1$ is the relative orientation of the shared link in the two plaquette boundaries (the incidence product), and the C-odd hop has no vacuum route. The C-odd channel-family split is $N_{\text{mixed}} = -\frac{27}{68}$ (channels $1, 8$), $N_{\text{like}} = -\frac{7}{18}$ (channels $\bar{3}, 6$), with sum $-\frac{481}{612}$ and difference $\frac{5}{612}$.

Remark 4.4 (Adjudication)**.** *Equation* (8) *corrects eq. (22) of [14] ($-9397/1020 \to -217/1020$ after assembly; see Theorem 5.5). The correction is adjudicated by the companion's own exact rational diagonalization of* (1) *on the two-plaquette domino, performed* blind *to the perturbative assembly: the C-even one-flux levels are $\{\frac{1769}{3060}, \frac{13}{20}\}$ at the relevant order, i.e.* $\operatorname{diag} \pm |t_+|$ *with* $\operatorname{diag} = \frac{1879}{3060}$*, forcing $|t_+| = \frac{110}{3060} = \frac{11}{306}$; the C-odd pair $\{\frac{31}{68}, \frac{17}{36}\}$ gives mean $\frac{71}{153} = \frac{1}{2} - \frac{11}{306}$ and half-splitting exactly $\frac{5}{612}$. Every constant in this section is additionally certified by a 251-gate exact-arithmetic computer proof (Appendix A).*

5 The exact band structure at second order

Assemble the one-flux sector at momentum k: a 3×3 Bloch problem over plaquette orientations. Let $A(k)$ and $\widetilde{N}(k)$ denote the unsigned and signed Bloch matrices of the 12-regular shared-link adjacency (signs $s = \sigma_p(b)\sigma_{p'}(b)$, the incidence product on the shared link). Then through $O(y^2)$

$$H_{\text{eff}}^{(+)}(k) = \Big[\tfrac{8}{3} - y + \tfrac{13}{20}y^2\Big]I + y^2\Big[12\big(-\tfrac{481}{612} + \tfrac{3}{4}\big)I + t_+ A(k)\Big] \;=\; \Big[\tfrac{8}{3} - y\Big]I + y^2\Big[\tfrac{223}{1020}I - \tfrac{11}{306}A(k)\Big], \tag{10}$$

$$H_{\text{eff}}^{(-)}(k) = \Big[\tfrac{8}{3} + y\Big]I + y^2\Big[\tfrac{7}{102}\,I + \tfrac{5}{612}\,\widetilde{N}(k)\Big]. \tag{11}$$

Theorem 5.1 (Incidence factorization; Gauss law)**.** *Let $N(k)$ and $B(k)$ be the unsigned and signed 3×3 Bloch incidence symbols of the oriented plaquette-boundary map (each plaquette feeds its four boundary links with incidence signs). With $u_j = 1 - e^{ik_j}$, $v_j = 1 + e^{ik_j}$,*

$$A(k) + 4I = N(k)N(k)^\dagger, \quad \det N(k) = -2v_1v_2v_3; \qquad \widetilde{N}(k) + 4I = B(k)B(k)^\dagger, \quad \det B(k) \equiv 0. \tag{12}$$

Hence $\lambda(k) \in [-4, 12]$ and $\mu(k) \in [-4, 8]$ for the spectra of A and $\widetilde{N}$, and the C-odd spectrum contains $\mu = -4$ at every *k: the lowest C-odd band is the kernel of $B(k)^\dagger$, the states with zero net signed amplitude into every link — a lattice Gauss law.*

Proof. B is the boundary symbol: row $(\mu\nu)$ has entries $e^{-ik_\nu/2} - e^{ik_\nu/2}$ and $e^{ik_\mu/2} - e^{-ik_\mu/2}$ in the link channels μ, ν (plaquette-center gauge) and 0 in the transverse channel; $BB^\dagger$ reproduces, entry by entry, the signed adjacency plus $4I$ (each plaquette has four boundary links), which is verified symbolically. Each row of B is a linear combination of two link channels, and the three rows involve the three channels cyclically with proportional weights; the 3×3 determinant cancels identically (a two-term cancellation $\propto \sin\frac{k_0}{2}\sin\frac{k_1}{2}\sin\frac{k_2}{2}$). The unsigned statement is the same computation without signs. □

Theorem 5.2 ($SU(N)$ shared-link universality)**.** *For every integer $N \geq 3$, let F denote the fundamental representation, $C_F = (N^2 - 1)/(2N)$, and use the reduced shared-link normalization*

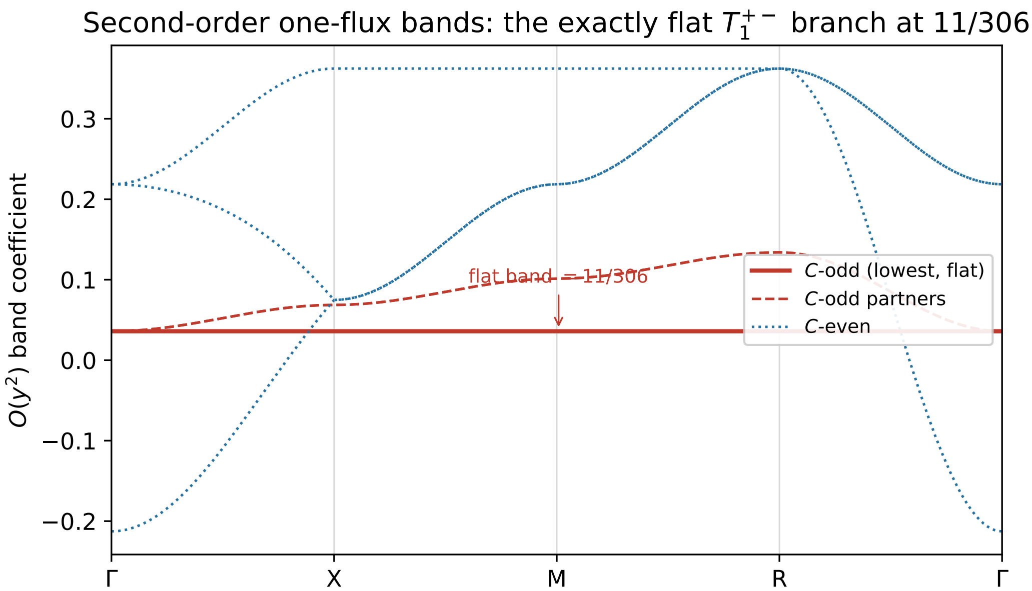

Figure 1: Second-order one-flux band coefficients (the $O(y^2)$ term) along Γ–X–M–R–Γ. The lowest C-odd branch (solid) is exactly flat at 11/306 for every momentum—the Gauss-law T_1^{+-} state of Theorem 5.3; its dispersive C-odd partners (dashed) touch it quadratically only at Γ. The C-even bands (dotted) disperse normally with width 88/153. Computed from the exact incidence symbols $\widetilde{N}(k)$ and $A(k)$ of Theorem 5.1.

in which the $SU(3)$ signed hop is $5/612$. The mixed-orientation fusion $F \otimes \bar{F} = 1 \oplus \mathrm{Adj}$ and the like-orientation fusion $F \otimes F = A_2 \oplus S_2$ give

$$w_1 = -\frac{2}{N(N-1)(N+1)}, \qquad w_{\mathrm{Adj}} = -\frac{2(N-1)(N+1)}{N(2N^2-1)},$$
$$w_{A_2} = -\frac{N-1}{(N+1)(2N-3)}, \qquad w_{S_2} = -\frac{N+1}{(N-1)(2N+3)}. \tag{13}$$

Consequently

$$W_{\mathrm{mixed}} = -\frac{2N^3}{(N^2-1)(2N^2-1)},$$
$$W_{\mathrm{like}} = -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)},$$
$$\Sigma_N := W_{\mathrm{like}} + W_{\mathrm{mixed}} = -\frac{2N(8N^4-19N^2+4)}{(N^2-1)(4N^2-9)(2N^2-1)},$$
$$t_N := W_{\mathrm{like}} - W_{\mathrm{mixed}} = \frac{2N(N^2-4)}{(N^2-1)(4N^2-9)(2N^2-1)} > 0. \tag{14}$$

The second-order C-odd operator therefore has the form $H_{2,N}(k) = d_N I + t_N \widetilde{N}(k)$ for a scalar d_N. Since $\widetilde{N}(k)$ has the exact eigenvalue -4 at every momentum, every $SU(N)$, $N \geq 3$, has a momentum-independent closed-surface eigenvalue $d_N - 4t_N$ at second order. Moreover,

$$\Sigma_3 = -\frac{481}{612}, \qquad t_3 = \frac{5}{612}, \qquad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} + O(N^{-7}).$$

Proof. After the six free-link integrations, the normalized shared-link tensor is isotropic on the N^2-dimensional product space. If P_R is the orthogonal projector onto an irreducible channel R, its squared channel norm is $\mathrm{Tr}\, P_R/N^2 = d_R/N^2$. The six nonshared fundamental links contribute $3C_F$ to the intermediate electric energy, while the fused shared link contributes $C_R/2$; hence

$$E_0 = 2C_F, \qquad E_R = 3C_F + \frac{C_R}{2}, \qquad w_R = -\frac{d_R/N^2}{C_F + C_R/2}.$$

Using

$$d_1 = 1,\ C_1 = 0; \quad d_{\mathrm{Adj}} = N^2 - 1,\ C_{\mathrm{Adj}} = N;$$

$$d_{A_2} = \frac{N(N-1)}{2},\ C_{A_2} = \frac{(N+1)(N-2)}{N}; \quad d_{S_2} = \frac{N(N+1)}{2},\ C_{S_2} = \frac{(N-1)(N+2)}{N}$$

gives (13). Summing the two orientation families and subtracting them gives (14). Every displayed factor in its numerator and denominator is positive for $N \geq 3$. The flat eigenvalue then follows from Theorem 5.1; the asymptotic expansion is direct. □

Theorem 5.3 (Exactly flat T_1^{+-} band; compact localized states)**.** *The lowest C-odd band is exactly flat:*

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306}\, y^2 + O(y^3) \qquad \textit{for every } k \in \mathrm{BZ}, \tag{15}$$

since $\frac{7}{102} + \frac{5}{612}(-4) = \frac{11}{306}$. The dispersive C-odd partners span coefficients $[\frac{11}{306}, \frac{41}{306}]$, touching the flat band quadratically at $k = 0$. At rest the C-odd triplet transforms as T_1^{+-} of the cubic group with $C = -$ (axial-vector-like): the C-odd one-plaquette sector contains no scalar at rest.

Theorem 5.4 (CLS, completeness, and $\partial\partial = 0$)**.** *The consistently (Levi–Civita) oriented boundary of a single elementary cube — six faces with amplitudes ± 1 — is an exact real-space eigenvector of the signed adjacency at $\mu = -4$ with zero leakage onto any plaquette off the cube. Its Bloch symbol $u(k)$ has $|u(k)|^2 = \sum_a \sin^2(k_a/2)$ and spans $\ker B(k)^\dagger$ for all $k \neq 0$. Flatness is the chain-complex identity $\partial_2\partial_3 = 0$: every edge of the cube is shared by exactly two faces with opposite induced orientation, so the signed face amplitudes cancel on all twelve edges. Exactly, on an L^3 torus,*

$$\dim\ker(\widetilde{N} + 4I) \;=\; L^3 + 2 \;=\; \underbrace{\big(L^3 \textit{ cube states},\ \textit{rank } L^3 - 1\big)}_{\textit{one global relation } \sum_{\mathrm{cubes}} \psi = 0} \;\oplus\; \underbrace{\big(3 \textit{ rest states}\big)}_{\widetilde{N}(0) = -4I}. \tag{16}$$

Theorem 5.5 (The C-even band)**.** *With the corrected hop* (8)*,*

$$E_+(k, y) = \frac{8}{3} - y + y^2\Big[\frac{223}{1020} - \frac{11}{306}\,\lambda(k)\Big], \qquad \lambda(k) \in [-4, 12], \tag{17}$$

with band bottom $-\frac{217}{1020}$ at $k = 0$ (the A_1^{++}, 0^{++} state), where $\lambda(k) = 12 - \frac{4}{3}|k|^2 + O(k^4)$ isotropically, hence curvature $+\frac{22}{459}|k|^2 y^2$ and effective mass $m^* = 459/(44\, y^2)$; band top $\frac{1109}{3060}$ at the zone faces; bandwidth $16|t_+| = \frac{88}{153}$. At rest the triplet decomposes as $A_1^{++} \oplus E^{++}$, with the E^{++} doublet at the exact, hop-independent coefficient $\frac{223}{1020}$ (it sits at $\lambda = 0$).

Remark 5.6 (An exactly immobile excitation)**.** *The per-bond ratio is*

$$\frac{|t_-|}{|t_+|} = \frac{5/612}{11/306} = \frac{5}{22},$$

and carries no immobility claim (this corrects Remark 6.4 of [14], whose ratio $5/481$ *used the uncorrected hop). The immobility statement holds in exact form: the lowest* C*-odd branch has bandwidth zero, while the full* C*-odd manifold has width* $\frac{5}{51}$ *against the* C*-even bandwidth* $\frac{88}{153}$ *(manifold-width ratio* $15/88$*). Both sectors factorize through the link channel; the qualitative asymmetry is that* $\det N = -2v_1v_2v_3$ *vanishes only on the zone faces* $k_j = \pi$*, while* $\det B \equiv 0$ *everywhere — the* C*-even kernel is measure-zero, the* C*-odd kernel is a whole band.*

6 Third order: tromino vanishing and a rigid flat band

At $O(y^3)$ the new lattice geometry is the tromino: three plaquettes pairwise sharing links.

Lemma 6.1 (Bare-link lemma)**.** *Every* $O(y^3)$ *tromino amplitude vanishes: each candidate three-plaquette process leaves at least one boundary link in a nontrivial representation unmatched ("bare"), and the corresponding Haar integral over that link is zero. Trominoes first activate at* $O(y^4)$*.*

Consequently the third-order effective Hamiltonian retains the second-order (signed-adjacency, link-mediated) structure. The C-odd hopping is odd in the shared-link orientation,

$$T_3(s) = -T_3(-s),$$

and the per-neighbor constants are computable on the abstract domino.

Theorem 6.2 (Rigid flat band at third order)**.** *The flat band of Theorem 5.3 is rigid at* $O(y^3)$*:*

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306}\, y^2 - \frac{109151}{249696}\, y^3 + O(y^4) \qquad \textit{for every } k, \tag{18}$$

with

$$d_3 = \frac{7}{32} + 12\,\mathrm{leak}_3 - 4\,b_3 = -\frac{109151}{249696}, \qquad b_3 = \frac{1975}{124848}, \quad \mathrm{leak}_3 = -\frac{12331}{249696}, \tag{19}$$

where $\frac{7}{32}$ *is the within-plaquette tower term* (5)*,* b_3 *the third-order* C*-odd hop and* leak_3 *the per-neighbor diagonal leakage; the dispersive* C*-odd top has cubic coefficient* $-\frac{61751}{249696}$*. In the* C*-even sector,*

$$m_+(0, y) = \frac{8}{3} - y - \frac{217}{1020}\, y^2 - \frac{54049}{520200}\, y^3 + O(y^4), \tag{20}$$

the band-top cubic coefficient is $\frac{471353}{1560600}$*, and the diagonal and hopping corrections obey the exact identity* $\mathrm{leak}^{\mathrm{e}}_n = T^{\mathrm{e}}_n$ *at* $n = 2, 3$ *(two distinct "+3/4" mechanisms of equal value), so the* A_1^{++} *correction at* $k = 0$ *is (tower)* $+24\,T^{\mathrm{e}}_n$ *per order. From the certified band form, the* E^{++} *level acquires the cubic coefficient* $\frac{101}{200} + 12\,T^{\mathrm{e}}_3 = \frac{52163}{260100}$ *and the* C*-even curvature the correction* $+\frac{4}{3}|T^{\mathrm{e}}_3|\, y^3 = \frac{6335}{187272}\, y^3$*.*

7 Robustness and exact fourth-order lifting

Theorem 7.1 (Link-mediated robustness)**.** *Let* $H_{\mathrm{corr}}(k) = B(k)M(k)B(k)^\dagger$ *be any correction whose symbol factors through the link channel, with* $M(k)$ *an arbitrary Hermitian symbol. Then* H_{corr} *annihilates the flat subspace:*

$$H_{\mathrm{corr}}(k)u(k) = B(k)M(k)\big(B(k)^\dagger u(k)\big) = 0.$$

Consequently every link-mediated correction preserves the flat branch. A separate k-independent scalar multiple of the identity may shift its energy without producing dispersion. This explains the exact second- and third-order rigidity.

Theorem 7.2 (Sharpness)**.** *The minimal corner-sharing, site-mediated symbol*

$$H_{\mathrm{corner}}(k) = \mathrm{diag}\big(2\cos(k_0+k_1), 2\cos(k_1+k_2), 2\cos(k_0+k_2)\big)$$

does not preserve the lower-order flat subspace: $P_{\mathrm{flat}} H_{\mathrm{corner}} P_\perp \neq 0$. Geometry alone does not protect the band after site-mediated processes activate.

7.1 The complete fourth-order kernel

Let the exact translation-invariant fourth-order kernel be $K_4(r)_{ba}$, where $a, b \in \{01, 02, 12\}$ label plaquette orientations and $r \in \mathbb{Z}^3$ is the relative cell displacement. The recovered kernel contains 189 nonzero rational records and obeys exact real-space Hermiticity,

$$K_4(r)_{ba} = K_4(-r)_{ab}.$$

Its Bloch matrix is

$$H_4(k) = \sum_r K_4(r) e^{ik\cdot r}.$$

At zero momentum the full matrix is scalar,

$$H_4(0) = qI_3, \qquad q = -\frac{20721577909065127111}{7250590288602460800}. \tag{21}$$

This is both an exact kernel sum and the cubic-symmetry requirement on the rest-frame T_1^{+-} triplet.

For $k \neq 0$ modulo reciprocal-lattice periodicity, use the cell/origin-gauge flat vector

$$\psi(k) = \begin{pmatrix} e^{ik_2} - 1 \\ -(e^{ik_1} - 1) \\ e^{ik_0} - 1 \end{pmatrix}, \qquad \|\psi(k)\|^2 = 2\sum_{i=0}^{2} X_i, \qquad X_i = 1 - \cos k_i. \tag{22}$$

It spans the one-dimensional lower-order flat eigenspace away from the threefold band touching at Γ.

Theorem 7.3 (Exact fourth-order factorization)**.** *Define*

$$D(k) = \psi(k)^\dagger [H_4(k) - qI] \psi(k).$$

The complete 189-record kernel gives the identity

$$D(k) = A\sum_i X_i^2 + B\sum_{i<j} X_i X_j, \qquad A = \frac{5}{12}, \qquad B = \frac{17607806155349}{275331901291200}. \tag{23}$$

Therefore the fourth-order coefficient of the branch that is flat through third order is

$$c_4(k) = q + \frac{A\sum_i X_i^2 + B\sum_{i<j} X_i X_j}{2\sum_i X_i}, \qquad k \neq \Gamma, \tag{24}$$

with continuous value $c_4(\Gamma) = q$.

Proof. Contracting the exact Laurent-polynomial kernel with (22) leaves 25 nonzero frequencies. Hermiticity and cubic symmetry reduce them to three cosine orbits: unit-axis frequencies, double-axis frequencies, and the six pair frequencies. Thus

$$D = C + a\sum_i \cos k_i + d\sum_i \cos 2k_i + b\sum_{i<j}\{\cos(k_i+k_j)+\cos(k_i-k_j)\}.$$

The exact kernel coefficients satisfy

$$C + 3a + 3d + 6b = 0, \qquad 2d = \frac{5}{12}, \qquad 2b = B, \qquad -a - 4d - 4b = 0.$$

The first identity is $D(0) = 0$; the last is the cancellation of the term linear in $X_i = 1 - \cos k_i$, equivalently the continuity of $D(k)/(2\sum_i X_i)$ with value zero at Γ. Substitution of $\cos 2k_i = 1 - 4X_i + 2X_i^2$ and $\cos(k_i+k_j)+\cos(k_i-k_j) = 2(1-X_i)(1-X_j)$ gives (23). Equation (24) follows from $\|\psi\|^2 = 2\sum_i X_i$. At Γ, (21) gives the continuous value directly. □

The complete strong-coupling branch through fourth order is consequently

$$m_-(k,y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 + c_4(k)y^4 + O(y^5). \tag{25}$$

Theorem 7.4 (Exact real-space sum of squares)**.** *Let $C = \partial_3$ map scalar cube amplitudes to their consistently oriented plaquette boundaries in the cell/origin gauge, so its Fourier symbol is $\psi(k)$ in (22). Let T_i translate cube amplitudes by one cell and define*

$$\nabla_i = T_i - I, \qquad L_i = \nabla_i^\dagger \nabla_i = 2I - T_i - T_i^{-1}.$$

Then the complete fourth-order kernel obeys the exact local operator identity

$$\boxed{C^\dagger(H_4 - qI)C = \frac{A}{4}\sum_i L_i^2 + \frac{B}{4}\sum_{i<j} L_i L_j} \tag{26}$$

with A, B from (23). Equivalently,

$$C^\dagger(H_4 - qI)C = \frac{5}{48}\sum_i L_i^2 + \frac{17607806155349}{1101327605164800}\sum_{i<j}(\nabla_i\nabla_j)^\dagger(\nabla_i\nabla_j). \tag{27}$$

The scalar numerator operator $Q = C^\dagger(H_4 - qI)C$ is the 25-point stencil

$$\begin{aligned}(Q\phi)_x = w_0\phi_x &+ w_1\sum_i(\phi_{x+e_i} + \phi_{x-e_i}) + w_2\sum_i(\phi_{x+2e_i} + \phi_{x-2e_i}) \\ &+ w_d\sum_{i<j}\sum_{\sigma,\tau=\pm1}\phi_{x+\sigma e_i+\tau e_j},\end{aligned} \tag{28}$$

where

$$\begin{aligned} w_0 &= \frac{189690244462349}{91777300430400}, \\ w_1 &= -\frac{132329431693349}{275331901291200}, \\ w_2 &= \frac{5}{48}, \qquad w_d = \frac{17607806155349}{1101327605164800}, \end{aligned} \tag{29}$$

and $w_0 + 6w_1 + 6w_2 + 12w_d = 0$.

The cube-boundary states have Gram operator

$$G = C^\dagger C = \sum_i L_i. \tag{30}$$

Consequently the physical lift is the generalized eigenproblem

$$Q\phi = \lambda G\phi, \qquad \lambda(k) = c_4(k) - q = \frac{A\sum_i X_i^2 + B\sum_{i<j} X_i X_j}{2\sum_i X_i}. \tag{31}$$

Proof. The symbol of L_i is $|e^{ik_i} - 1|^2 = 2X_i$. Hence the Fourier symbol of the right side of (26) is

$$\frac{A}{4}\sum_i (2X_i)^2 + \frac{B}{4}\sum_{i<j}(2X_i)(2X_j) = A\sum_i X_i^2 + B\sum_{i<j} X_i X_j,$$

which equals the exact contracted symbol $D(k)$ by Theorem 7.3. Since both sides are finite-range translation-invariant operators, equality of their Laurent symbols proves the real-space identity. Expanding L_i^2 and L_iL_j gives (28)–(29). Equation (30) follows from the definition of C, and division by its symbol $2\sum_i X_i$ gives (31). The exact certificate also contracts the 189-record kernel directly and obtains the same 25 rational coefficients with zero residual. $\square$

Remark 7.5 (Geometric carriers of fourth-order mobility)**.** *For a localized cube amplitude, L_i creates the axial three-cube pattern $(-1, 2, -1)$, while $\nabla_i\nabla_j$ creates a planar four-cube checkerboard $(+1, -1, -1, +1)$. Equation (27) states that the first mobility of the closed-surface excitation is exactly the positive weighted norm of these two local deformations.*

Theorem 7.6 (Exact global band edges)**.** *Modulo reciprocal-lattice periodicity, $\Gamma = (0, 0, 0)$ is the unique global minimum of $c_4(k)$ and $R = (\pi, \pi, \pi)$ is the unique global maximum. The exact fourth-order bandwidth is*

$$\Delta W_4 = c_4(R) - c_4(\Gamma) = A + B = \frac{132329431693349}{275331901291200}. \tag{32}$$

Proof. For $k \neq \Gamma$, the variables satisfy $X_i \in [0, 2]$ and $S = \sum_i X_i > 0$. Since $A > 0$, $B > 0$, and $\sum_i X_i^2 > 0$, equation (23) gives $D(k) > 0$. Hence $c_4(k) > q = c_4(\Gamma)$.

For the upper edge, write $Q = \sum_i X_i^2$ and $R_2 = \sum_{i<j} X_i X_j$. On the cube $0 \le X_i \le 2$,

$$Q \le 2S, \qquad R_2 \le 2S.$$

For the second inequality set $x_i = X_i/2$; then

$$(x_0 + x_1 + x_2) - (x_0x_1 + x_0x_2 + x_1x_2) = x_0(1 - x_1) + x_1(1 - x_2) + x_2(1 - x_0) \ge 0.$$

Therefore $c_4(k) - q \le A + B$. Equality away from Γ requires $x_0 = x_1 = x_2 = 1$, hence $X_0 = X_1 = X_2 = 2$, which is precisely $R = (\pi, \pi, \pi)$ modulo 2π. $\square$

Proposition 7.7 (Kinematic generic-N factorization)**.** *Fix an integer $N \ge 3$ and define*

$$D_N(k) = \psi(k)^\dagger[H_{4,N}(k) - q_N I]\psi(k).$$

Suppose the exact rank-N kernel passes the group-independent gates of Hermiticity, cubic covariance, independent-reflection symmetry, the same final displacement support as the $SU(3)$ kernel, and $H_{4,N}(0) = q_N I$. Then

$$D_N(k) = A_N \sum_i X_i^2 + B_N \sum_{i<j} X_i X_j.$$

Writing q_N, c_X, c_M, c_R for the exact fourth-order branch coefficients at Γ, X, M, R, respectively,

$$A_N = c_X - q_N, \qquad B_N = 2(c_M - c_X) = c_R - c_X, \qquad c_R = 2c_M - c_X.$$

The last equality is a hard consistency gate on any generic-N computation. The proposition is purely kinematic. Theorem 7.11 verifies its hypotheses and the signs of A_N, B_N throughout the stable range $N \geq 7$, while Theorem 7.13 closes the band shape at $N = 3, 4, 5, 6$.

Proof. The final displacement support and cubic/reflection symmetries restrict D_N to a constant, a term proportional to $S = \sum_i X_i$, and the two quadratic cubic invariants $Q = \sum_i X_i^2$ and $R_2 = \sum_{i<j} X_i X_j$. Since $\psi(k) = O(k)$ and $H_{4,N}(k) - q_N I = O(k)$, analyticity gives $D_N(k) = O(|k|^3)$. Independent reflections make D_N even, hence in fact $D_N(k) = O(|k|^4)$. The constant and quadratic Taylor terms therefore vanish, leaving $A_N Q + B_N R_2$. Evaluating $(A_N Q + B_N R_2)/(2S)$ at X, M, and R gives A_N, $A_N + B_N/2$, and $A_N + B_N$, respectively, yielding the stated reconstruction identities. $\square$

The high-symmetry values are

$$\begin{aligned} c_4(\Gamma) &= -\frac{20721577909065127111}{7250590288602460800}, \\ c_4(X) &= -\frac{17700498622147435111}{7250590288602460800}, \\ c_4(M) &= -\frac{4367164159624988707}{1812647572150615200}, \\ c_4(R) &= -\frac{3447362930970494909}{1450118057720492160}. \end{aligned} \tag{33}$$

Proposition 7.8 (Band-edge regularity and curvature)**.** *For $k = tn$ with $|n| = 1$,*

$$c_4(tn) - q = \frac{t^2}{4}\left[A \sum_i n_i^4 + B \sum_{i<j} n_i^2 n_j^2\right] + O(t^4). \tag{34}$$

The directional coefficient ranges from

$$\kappa_{\text{diag}} = \frac{A+B}{12} = \frac{132329431693349}{3303982815494400}$$

on a body diagonal to

$$\kappa_{\text{axis}} = \frac{A}{4} = \frac{5}{48}$$

on a coordinate axis. Thus the Γ lift is quadratic but not the restriction of a single Hessian. At the upper edge,

$$c_4(R+\delta) = c_4(R) - \frac{132329431693349}{3303982815494400}|\delta|^2 + O(|\delta|^4), \tag{35}$$

so the R curvature is isotropic and negative.

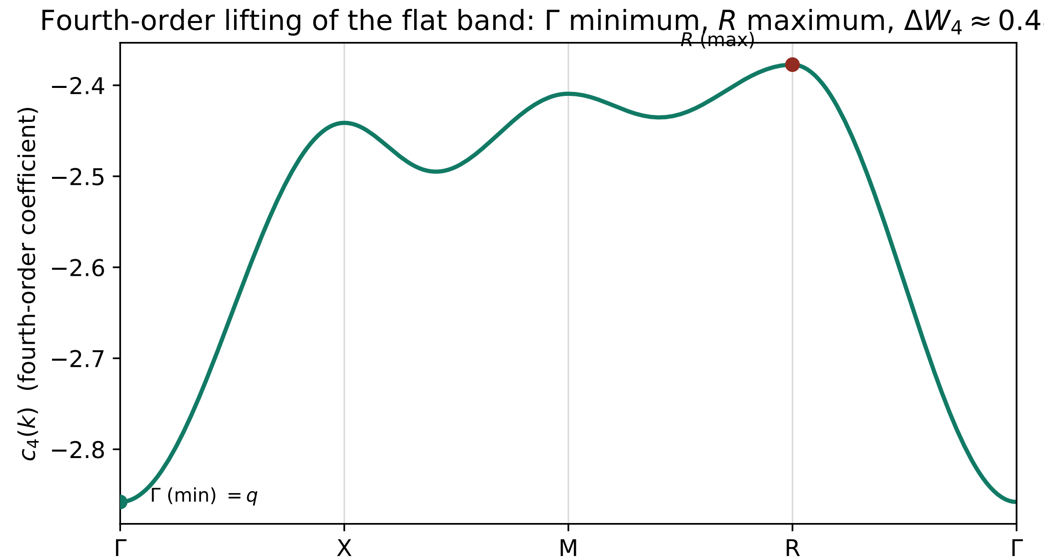

Figure 2: The fourth-order coefficient $c_4(k)$ of equation (24) along Γ–X–M–R–Γ, exhibiting the exact global structure of Theorem 7.6: a unique minimum at Γ (value q), a unique maximum at R, and the parameter-free bandwidth $\Delta W_4 = A+B \approx 0.4806$. This is the first nonzero dispersion of the closed-surface branch, exactly immobile through $O(y^3)$.

Proof. Near Γ, $X_i = t^2 n_i^2/2 + O(t^4)$; insertion into (24) gives (34). Since $\sum_i n_i^4 \in [1/3, 1]$, $\sum_{i<j} n_i^2 n_j^2 = (1 - \sum_i n_i^4)/2$, and $A - B/2 > 0$, the stated extrema follow. If a Hessian existed at Γ, cubic symmetry would force it to be a scalar multiple of the identity, contradicting the two distinct directional coefficients. Near R, $X_i = 2 - \delta_i^2/2 + O(\delta_i^4)$; expanding the ratio in (24) gives (35). □

7.2 Stable-rank $SU(N \geq 7)$ fourth-order reduction

Theorem 7.9 (Stable-rank local-invariant and denominator theorem)**.** *For the complete connected fourth-order geometry, the maximum local tensor degree is six. For every integer $N \geq 7$, all local $SU(N)$ singlets are therefore exactly balanced in fundamental and antifundamental degree; no independent determinant/epsilon sectors occur. The local Haar integrations reduce to the three balanced families* $(1,1)$, $(2,2)$, *and* $(3,3)$, *with*

$$\mathrm{Wg}_1(e) = \frac{1}{N},$$

$$\mathrm{Wg}_2(e) = \frac{1}{N^2-1}, \qquad \mathrm{Wg}_2((12)) = -\frac{1}{N(N^2-1)},$$

and

$$\mathrm{Wg}_3(e) = \frac{N^2-2}{N(N^2-1)(N^2-4)},$$

$$\mathrm{Wg}_3((12)) = -\frac{1}{(N^2-1)(N^2-4)}, \qquad \mathrm{Wg}_3((123)) = \frac{2}{N(N^2-1)(N^2-4)}.$$

The stable mixed-tensor irreducible channels are indexed by bipartitions (λ, μ), with

$$C_2(\lambda, \mu) = \frac{N^2(|\lambda| + |\mu|) + N[\kappa(\lambda) + \kappa(\mu)] - (|\lambda| - |\mu|)^2}{2N}, \qquad \kappa(\lambda) = \sum_i \lambda_i(\lambda_i + 1 - 2i).$$

Consequently every electric denominator is an explicit rational function of N. Exact enumeration yields the counts in Table 1; there are 94 distinct denominator polynomials and none has an accidental integer root for $N \geq 7$. Evaluating the stable bipartition engine at $N = 3$ agrees with the existing $SU(3)$ Dynkin-label engine on all 140 balanced token signatures.

Proof. A local invariant with n_f fundamental and $n_{\bar{f}}$ antifundamental indices requires $n_f - n_{\bar{f}} \equiv 0 \pmod N$. The verified local degree bound gives $|n_f - n_{\bar{f}}| \leq 6$; for $N \geq 7$ this congruence is equivalent to $n_f = n_{\bar{f}}$. The Weingarten coefficients are the exact inverses of the permutation Gram matrices through degree three. Stable mixed-tensor branching then gives the displayed bipartition Casimir. The geometry counts, denominator classification, absence of integer roots, and 140-signature regression are exact integer and symbolic gates in the certificate described in Appendix A. □

Theorem 7.10 (Exact stable-rank local walled-Brauer projectors)**.** *For every compact balanced token family of local degree $r \in \{1, 2, 3\}$, let the invariant pairing basis be indexed by $\sigma \in S_r$ and let*

$$G_{\sigma\tau} = N^{\#\,\mathrm{cycles}(\sigma^{-1}\tau)}.$$

For each mixed-tensor branching path p, there is an exact rank-one action matrix Q_p over $\mathbb{Q}(N)$. In the $r!$-dimensional pairing basis, its channel-resolved Haar dyad is

$$D_p = Q_p G^{-1}.$$

The complete set satisfies

$$Q_p^2 = Q_p, \qquad \operatorname{tr} Q_p = 1, \qquad Q_p^T G = G Q_p, \qquad C_{\mathrm{total}} Q_p = 0,$$

$$Q_p Q_{p'} = 0 \quad (p \neq p'), \qquad \sum_p Q_p = I, \qquad \sum_p D_p = G^{-1}, \qquad D_p^T = D_p.$$

Exact enumeration gives 28 compact token families, 134 distinct path projectors, 730 spectral factors, and 330 path instances across the 140 balanced six-event signatures. All 746 Weingarten-entry gates, 1,338 projector-matrix gates, and 140 Stage-1 channel-history regressions pass. Consequently the local Stage-3C carrier/projector layer is closed over $\mathbb{Q}(N)$ for every integer $N \geq 7$.

Proof. Nested partial-Casimir spectral factors construct the path action matrices in the pairing basis. Direct symbolic reduction over $\mathbb{Q}(N)$ verifies the displayed projector, Gram-adjoint, orthogonality, completeness, and total-Casimir identities. Multiplication by the exact Weingarten inverse gives D_p and the path sum $\sum_p D_p = G^{-1}$. The complete serialized matrices and the independent gate re-evaluation are described in Appendix A. □

Theorem 7.11 (Stable-rank full fourth-order coefficient theorem)**.** *For every integer $N \geq 7$, the complete global Stage-3G trace-index contraction passes the kernel hypotheses of Proposition 7.7. With $X_i = 1 - \cos k_i$,*

$$D_N(k) = \psi(k)^\dagger [H_{4,N}(k) - q_N I]\psi(k) = A_N \sum_i X_i^2 + B_N \sum_{i<j} X_i X_j,$$

quantity	count	quantity	count
connected supports	182440	candidate support/output pairs	895524
stable support/output classes	439	canonical ordered words	4171
exact-balance sign assignments	33500	charge-conjugation orbits	16750
unique balanced signatures	140	symbolic energy signatures	37500
resonant signatures	17073	nonresonant signatures	20427
distinct denominator polynomials	94	accidental roots, $N \geq 7$	0
compact token families	28	unique path projectors	134
six-event path instances	330	spectral factors	730
Weingarten-entry gates	746	projector-matrix gates	1338
trace topologies	3850	global fusion paths	35130
q_N topologies / paths	2290 / 27202	q_N denominator classes	167
A_N topologies / paths	406 / 950	A_N energy groups	140
B_N topologies / paths	2328 / 13096	B_N energy groups	743

Table 1: Exact stable-rank fourth-order geometry, local-projector, and global-contraction counts.

and, for $k \neq \Gamma$,

$$c_{4,N}(k) = q_N + \frac{A_N \sum_i X_i^2 + B_N \sum_{i<j} X_i X_j}{2 \sum_i X_i}.$$

Set $z = N^2$. The exact coefficients are

$$q_N = -\frac{2}{3N}\frac{Q_{32}(z)}{D_{34}(z)}, \qquad A_N = \frac{640}{N(N^2-1)^3}, \qquad B_N = \frac{P_{17}(z)}{N R_{20}(z)},$$

where Q_{32}, D_{34} are the exact integer polynomials in the supplementary q_N ledger and P_{17}, R_{20} are displayed in Appendix A.1. For every integer $N \geq 7$,

$$q_N < 0, \qquad A_N > 0, \qquad B_N > 0.$$

Modulo reciprocal-lattice periodicity, Γ is therefore the unique global minimum of $c_{4,N}$, $R = (\pi,\pi,\pi)$ is the unique global maximum, and

$$\boxed{\Delta c_{4,N} = c_{4,N}(R) - c_{4,N}(\Gamma) = A_N + B_N > 0.}$$

The exact contraction census is 3,850 trace topologies and 35,130 global fusion paths. The q_N, A_N, and B_N sectors receive respectively 27,202, 950, and 13,096 nonzero path contributions.

Proof. The local projectors of Theorem 7.10 are contracted through the oriented trace boundaries of all 4,171 ordered words by exact partition/loop elimination. The resulting fixed-rank kernels have 189 real-space entries, satisfy Hermiticity and cubic covariance, obey $H_{4,N}(0) = q_N I$, and give identical values of $2[c_{4,N}(M) - c_{4,N}(X)]$ and $c_{4,N}(R) - c_{4,N}(X)$. Thus Proposition 7.7 yields the displayed two-invariant form.

For q_N, every factor of $D_{34}(z)$ is positive for $z = N^2 \geq 49$, while

$$Q_{32}(z) = \sum_{j=0}^{32} b_j \binom{z-49}{j}, \qquad b_j > 0.$$

Hence $q_N < 0$. The sign of A_N is immediate. The original 74-term structured expression for B_N reduces exactly to the displayed compact form after cancellation of $(4N-5)(4N+5)(4N^2-3)$. Every factor of $R_{20}(z)$ is positive for $z \geq 49$, and

$$P_{17}(z) = \sum_{j=0}^{17} a_j \binom{z-49}{j}, \qquad a_j > 0,$$

so $B_N > 0$. Exact fixed-rank contractions for $N = 7, \ldots, 18$, including the unused $N = 18$ holdout, match the symbolic formulas. The global extrema and bandwidth then follow from the cube inequalities used in Theorem 7.6. $\square$

Corollary 7.12 (Stable-rank large-N collapse)**.** *The exact compact forms give*

$$q_N = -\frac{227}{N^5} + O(N^{-7}), \qquad A_N = \frac{640}{N^7} + O(N^{-9}), \qquad B_N = \frac{6170}{9N^7} + O(N^{-9}).$$

Consequently

$$\frac{B_N}{A_N} \longrightarrow \frac{617}{576}, \qquad \Delta c_{4,N} = \frac{11930}{9N^7} + O(N^{-9}).$$

Theorem 7.13 (Fourth-order band shape for all non-pseudoreal ranks)**.** *For $SU(3)$,*

$$A_3 = \frac{5}{12}, \qquad B_3 = \frac{17607806155349}{275331901291200}.$$

For every integer $N \geq 4$,

$$A_N = \frac{640}{N(N^2-1)^3}, \qquad B_N = \frac{P_{17}(N^2)}{N R_{20}(N^2)}.$$

For every integer $N \geq 3$,

$$A_N > 0, \qquad B_N > 0.$$

Therefore Γ is the unique global minimum, R is the unique global maximum, and

$$\boxed{\Delta c_{4,N} = A_N + B_N > 0}$$

for every non-pseudoreal $SU(N)$.

Proof. The case $N = 3$ is Theorem 7.3. For $N \geq 7$ the statement is Theorem 7.11. A fresh exact N-ality scan gives the following finite exceptional-rank reductions. At $N = 4$ there are 312 additional determinant-sector assignments, or 156 charge-conjugation orbits. Only 40 exceptional ordered words enter the 64-key A/B target. Their contraction contains 48 active trace topologies in nine equal-amplitude classes: every A-functional coefficient vanishes individually, while each B class has equal multiplicities of coefficients $+8$ and -8. Hence $\Delta A_4 = \Delta B_4 = 0$. At $N = 5$ no determinant assignment occurs anywhere in the complete fourth-order geometry. At $N = 6$ one determinant charge-conjugation orbit occurs, but it is absent from all 64 entries defining A_6 and B_6. Thus the balanced compact formulas hold at $N = 4, 5, 6$. Their positivity is checked by exact substitution, completing the proof. $\square$

Theorem 7.14 ($SU(2)$ exclusion)**.** *In pure $SU(2)$ gauge theory there is no one-flux T_1^{+-} branch. Every fundamental matrix obeys $U^* = \varepsilon U \varepsilon^{-1}$ with $\varepsilon = i\sigma_2$; the constant gauge transformation $g_x = \varepsilon$ at every vertex sends $U_{xy} \mapsto U_{xy}^*$, so charge conjugation is a gauge transformation and acts trivially on the physical Hilbert space. Hence $C = I$, $P_{C=-} = 0$, and q_2, A_2, B_2 are undefined. The band-shape theorem domain $N \geq 3$ is therefore maximal.*

Remark 7.15 (Determinant offsets at exceptional rank). *The common offset q_3 is known from the rank-specific $SU(3)$ kernel, and q_5 (fourth order, $SU(5)$) is given by the balanced formula in Theorem 7.11. The determinant-sector corrections to the remaining exceptional offsets are* $\Delta q_4 = -304746539168/160249753125$ *and* $\Delta q_6 = 6/343$*, both scalar on the closed-surface branch with* $\Delta A = \Delta B = 0$ *($SU(5)$ has no determinant sector). They shift only the rest energy and do not enter the extrema or bandwidth theorem.*

8 SU(3) fifth order and the scale-matched mass ratio

Everything above is fourth order and holds for all ranks $N \geq 3$. The same $SU(3)$ machinery extends to fifth order in y. The fifth-order kernel is again a 189-record real-space object, and the projected coefficient has the *identical* two-invariant form as fourth order.

Theorem 8.1 (Fifth-order $SU(3)$ band). *For $X_i = 1 - \cos k_i$, $S = \sum_i X_i$, $Q = \sum_i X_i^2$, $R = \sum_{i<j} X_i X_j$,*

$$c_5(k) = q_5 + \frac{A_5\, Q + B_5\, R}{2S}, \qquad A_5 = \frac{313}{240}, \quad B_5 = \frac{188186308774290860590 3793}{165293224897596718104 0000},$$

$$q_5 = -\frac{86623675050334202625309669 1057}{116966808379381140344713348 8000} \approx -0.740583386437.$$

Both shape coefficients are positive ($A_5 \approx 1.3042$, $B_5 \approx 1.1385$), so Γ is the unique minimum and R the unique maximum of c_5, with exact bandwidth

$$\Delta c_5 = A_5 + B_5 = \frac{403756222911573247117679 3}{165293224897596718104 0000} \approx 2.442666.$$

The parity anchors satisfy the same exact consistency gate $c_R = 2c_M - c_X$ and the extraction relations $A_5 = c_X - q_5$, $B_5 = 2(c_M - c_X) = c_R - c_X$, both checked to agree. The flat-band picture therefore persists through $O(y^5)$; what changes is only magnitude—the fifth-order shape coefficients are $O(1)$, so the band is no longer approximately flat at this order. The complete rest-mass series is

$$m_{1^{+-}}(y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 - \frac{207215779090651271 11}{725059028860246080 0}y^4 - \frac{86623675050334202625309669 1057}{116966808379381140344713348 8000}y^5 + O(y^6). \tag{36}$$

The fifth-order contraction is substantially larger than fourth: 6,676,658 connected five-insertion supports, 29,366 canonical ordered words, 524,823 global path contractions, and 189 nonzero kernel records; the new local sectors are $(4,1)$ and $(5,2)$.

8.1 Project-native string tension and the scale-matched ratio

The static-flux (torelon) strong-coupling expansion gives the Hamiltonian string tension in the same variable $y = \beta/6$. The connected-support engine reproduces, L-independently and exact against the Kogut–Pearson–Shigemitsu (KPS) table through fourth order,

$$\sigma(y) = \frac{2}{3} - \frac{22}{153}y^2 - \frac{61}{408}y^3 - \frac{737327120374220449}{725059028860246080 0}y^4 + O(y^5).$$

The conversion to the KPS variable $x = 2y$ is the single bridge $\sigma(y) = \frac{1}{2}W(2y)$; this sign convention—in particular the *negative* odd coefficient $\sigma_3 = -61/408$—is forced by the native engine (the alternative $\frac{1}{2}W(-2y)$ gives $+61/408$ and is excluded). The fifth-order tension σ_5 has since been reproduced by an independent *native* first-principles computation (Appendix B): an exact $SU(3)$ torelon strong-coupling engine that, unlike the adjoint-sector contraction, resolves the determinant/ε (triality) sector that carries all odd orders. It reproduces σ_2, σ_3 in exact rational arithmetic and now determines σ_5 in *exact finite-field* arithmetic: computed modulo seven independent primes, CRT-combined (189-bit modulus), and rational-reconstructed with no literature input (Appendix B), coinciding with the historical KPS value. Thus σ_5 is exact, not merely a KPS target. The sixth tension σ_6 remains an exact KPS historical target (its native evaluation is of larger, external-memory scale). The scale-matched ratio is then

$$\frac{m_{1+-}(y)}{\sqrt{\sigma(y)}} = \sqrt{6}\sum_{n=0}^{5} c_n y^n + O(y^6), \qquad \begin{array}{l} c_0 = \frac{4}{3}, \quad c_1 = \frac{1}{2}, \quad c_2 = \frac{11}{68}, \quad c_3 = -\frac{7559}{499392}, \\ c_4 = -\frac{15752822901180179}{12642703205932800}, \quad c_5 = -\frac{10670728893034386567182468628311}{46786723351752456137885339520000}, \end{array} \tag{37}$$

re-derived independently from the rest-mass and string-tension series. The sixth-order ratio coefficient is fixed up to one unknown, $c_6 = \frac{1}{2}m_6 + K$ with K exact, so only the sixth-order rest mass m_6 is missing to reach $O(y^6)$.

Remark 8.2 (Sixth order: a provisional internal value)**.** *The global* Γ*-contraction has since been carried out as an exact external-memory computation over the full* 247,326,161*-support census (*3,094,806 *canonical ordered words,* 205,699 *nonzero* Γ*-point topology blocks,* 10,907,384 *global local-path choices), summed as* 229 *exact rational shards. In the project convention it gives*

$$m_6 = -\frac{156998370765216917515896262601525405897211506214753116643443873}{488068179127562905075926479809565202795087879471974400000} \approx -32167.30,$$

and correspondingly, for the ratio $m_{1+-}/\sqrt{\sigma} = \sqrt{6}\sum_n c_n u^n$*,*

$$c_6 = -\frac{42703538244288992007864271915574871272499717015686613641478 01}{265509089445394220361304005016403470320527806432754073600} \approx -16083.64.$$

This is exact within the enumerated one-flux des-Cloizeaux construction*. Three internal gates support it: the identical* Γ*-reduction reproduces* q_5 *exactly; the* 32 *resonance-pattern folded weights agree with an independent Rayleigh–Schrödinger recurrence; and the dominant high-multiplicity anchor (folded weight* $-243/16384$*,* Γ*-multiplicity* 65208*, eight blocks summing to* $-1980693/256$*) is reproduced here independently by the folded-weight module used for the* σ_5 *certificate, which matches the contraction's fold formula on every test vector. An independent reimplementation now confirms the computational primitives: a weight-blocked Casimir-nullspace Haar engine reproduces the local singlet projectors* matrix-exactly *for all order-four signatures and matches the singlet dimension and cumulative-Casimir histories for every triality family through degree eight (the order-six carriers, including the determinant sectors); the* σ_5*-certified des-Cloizeaux module reproduces the contraction's folded weights exactly; and the full order-four pipeline was rerun from scratch, regenerating* q_4*. The result is nonetheless retained as* ***provisional****: what remains for theorem-level promotion is an independent re-derivation of the order-six* geometry census *(the* 247,326,161*-support enumeration and its* Γ *multiplicities), which is external-memory scale. The coefficient is anomalously large relative to* $q_2, \ldots, q_5$*; the audit traces this to a small family of high-multiplicity, low-path-dimension* Γ *blocks rather than to numerical instability, and the magnitude lives in those order-six multiplicities, not in the (now independently verified) color or fold layers.*

9 Numerical summary and discussion

Table 2 collects the exact lower-order data. These entries are unchanged from the third-order certificate chain.

quantity	exact	decimal
C-odd flat branch, $O(y)$ / $O(y^2)$ / $O(y^3)$	$+1$; $+11/306$; $-109151/249696$	1; 0.0359477; -0.4371356
C-odd dispersive top, $O(y^2)$ / $O(y^3)$	$41/306$; $-61751/249696$	0.1339869; -0.2473048
C-odd manifold width (y^2)	$5/51$	0.0980392
A_1^{++} bottom, $O(y)$ / $O(y^2)$ / $O(y^3)$	-1; $-217/1020$; $-54049/520200$	-1; -0.2127451; -0.1039004
E^{++} level, $O(y^2)$ / $O(y^3)$	$223/1020$; $52163/260100$	0.2186275; 0.2005498
C-even bandwidth (y^2)	$88/153$	0.5751634
C-even effective mass	$459/(44y^2)$	$10.43182/y^2$
per-bond ratio $\|t_-\|/\|t_+\|$	$5/22$	0.2272727

Table 2: Exact one-flux data through third order. The lowest C-odd branch is flat at every momentum through the displayed orders.

The new fourth-order data are summarized in Table 3.

quantity	exact	decimal
$c_4(\Gamma)$	$-20721577909065127111/7250590288602460800$	-2.857915988115
$c_4(X)$	$-17700498622147435111/7250590288602460800$	-2.441249321448
$c_4(M)$	$-4367164159624988707/1812647572150615200$	-2.409273720232
$c_4(R)$	$-3447362930970494909/1450118057720492160$	-2.377298119016
ΔW_4	$132329431693349/275331901291200$	0.480617869098
Γ body-diagonal coefficient	$132329431693349/3303982815494400$	0.040051489092
Γ axis coefficient	$5/48$	0.104166666667
R isotropic coefficient	$-132329431693349/3303982815494400$	-0.040051489092

Table 3: Exact projected $O(y^4)$ band data. The signs in the last row refer to the coefficient multiplying $|\delta|^2$ in the expansion about R.

Several consequences follow.

First, the Gauss-law mechanism is exact but order-limited. The elementary-cube CLS annihilate all link-mediated leakage through third order. Fourth-order site-mediated processes do not merely perturb this conclusion numerically: they produce the strictly positive form (23). The first nonzero bandwidth of the lowest T_1^{+-} branch is therefore

$$W(y) = \Delta W_4 y^4 + O(y^5).$$

The excitation is parametrically low-mobility at strong coupling, not all-orders immobile.

Second, the global band geometry is fixed without interval arithmetic. The minimum is at zero momentum and the maximum is at the Brillouin-zone corner. The rest-point degeneracy leaves a nonstandard signature: although $c_4(k) - c_4(\Gamma) = O(|k|^2)$, the coefficient depends on the approach direction. A conventional scalar effective mass at Γ is therefore not defined at this order. The upper edge at R has an ordinary isotropic negative curvature.

Third, the exact rest-frame series is

$$m_-(\Gamma, y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 - \frac{20721577909065127111}{7250590288602460800}y^4 + O(y^5). \tag{38}$$

The magnitude of the fourth-order coefficient is already large, so naive polynomial extrapolation to moderate coupling is not controlled. A quantitative comparison with the continuum 1^{+-} mass additionally requires the string tension in the same Hamiltonian normalization.

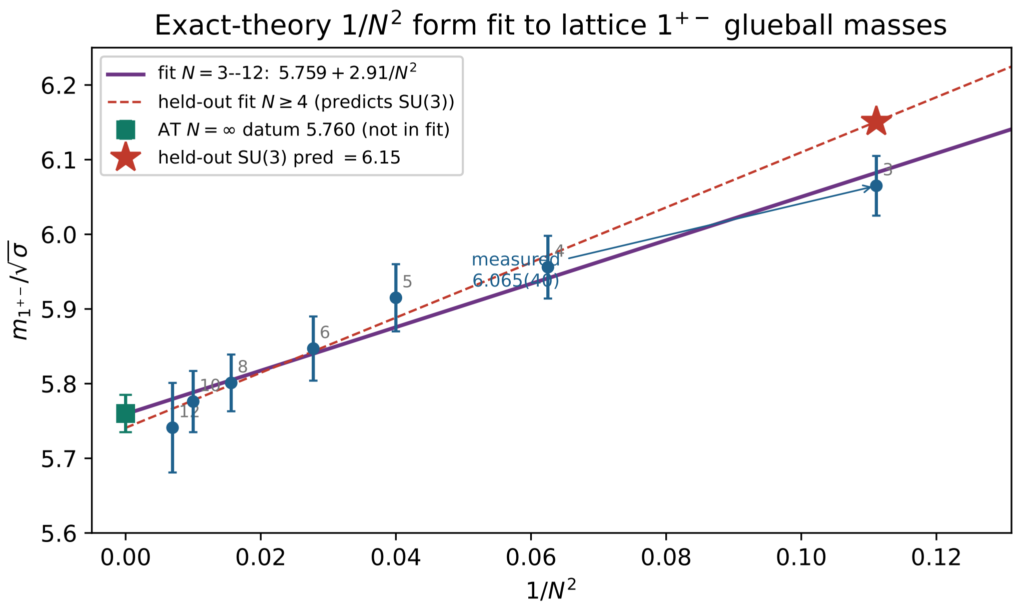

Figure 3: The exact-theory $1/N^2$ band-shape result fixes the functional form of the leading large-N correction to $m_{1^{+-}}/\sqrt{\sigma}$. Fitting $m_\infty + c/N^2$ to the Athenodorou–Teper continuum data [6] at $N = 3, \ldots, 12$ (solid) gives $m_\infty = 5.759(25)$, whose intercept coincides with the independently measured $N = \infty$ datum 5.760(25) (green square, not used in the fit). The held-out fit to $N \geq 4$ (dashed) predicts the physical $SU(3)$ ratio ≈ 6.15 (star) against the measured 6.065(40). This is an extrapolation of lattice data made trustworthy by the exact $1/N^2$ form, not a continuum mass from the strong-coupling series.

Fourth, the quantum numbers remain useful for spectroscopy. The lowest C-odd one-flux excitation is T_1^{+-}, hence 1^{+-}-like at rest, not a scalar. The lower-order ordering $0^{++} < 2^{++} < 1^{+-}$ agrees qualitatively with continuum lattice spectroscopy [5, 6], but the present series does not predict the continuum mass.

Finally, the result provides both momentum- and real-space benchmarks for Hamiltonian lattice simulations and quantum hardware [13]. At small y, a correct implementation should reproduce zero dispersion through $O(y^3)$, followed by a bandwidth $\Delta W_4 y^4$, with the exact Γ, X, M, and R coefficients in Table 3. In real space, sandwiching the full 189-record kernel between cube-boundary maps must collapse to the 25-point stencil (28). This paired test is substantially more discriminating than a single rest-mass comparison.

The all-rank result supplies a parallel gauge-group benchmark. For every integer $N \geq 3$, projected band differences must reproduce the exact A_N, B_N of Theorem 7.13, including the hard relation $c_{4,N}(R) = 2c_{4,N}(M) - c_{4,N}(X)$ and the strictly positive bandwidth $A_N + B_N$. For $N \geq 7$ (and also $N = 5$), the common shift must additionally reproduce q_N from Theorem 7.11. This tests the full color-contraction layer rather than only the $SU(3)$ spatial kernel.

10 Physical interpretation: the continuum 1^{+-} mass

Two facts frame the contact with continuum spectroscopy.

The strong-coupling ratio alone does not reach the continuum. With the scale-matched series (37) the strong-coupling anchor is $R(0) = \frac{4}{3}\sqrt{6} \approx 3.266$, while the continuum target is $m_{1+-}/\sqrt{\sigma} = 6.065(40)$ [6]. The fourth-order truncation has a unique positive stationary point at $y_* \approx 0.469$ with $R_4(y_*) \approx 3.730$—still 38.5% below the continuum—and $R_4(y) = 6.065$ has no positive real solution. Near-diagonal Padé, dlog-Padé, and Borel–Padé approximants are all defective on the positive axis or tend to the wrong limit. A controlled continuum value from the series therefore needs at minimum the fifth- and sixth-order ratio coefficients (hence m_6 and native σ_5, σ_6), plus finite-coupling anchors and the shell-six channel mixing.

The exact $1/N^2$ band structure does enable a controlled prediction across rank. The all-rank theorem fixes the leading large-N correction to the ratio to be $O(1/N^2)$. Fitting $m_{1+-}/\sqrt{\sigma}(N) = m_\infty + c/N^2$ to the Athenodorou–Teper continuum data at $N = 3, \ldots, 12$ [6] gives $m_\infty = 5.759(25)$ and $c = 2.91(46)$. The fitted $N \to \infty$ intercept matches the independently measured large-N datum 5.760(25)—not used in the fit—to 0.02σ, and leave-one-out validation gives held-out RMS 0.045, comparable to the lattice error bars. Predicting the held-out physical rank from $N \geq 4$ gives

$$\left.\frac{m_{1+-}}{\sqrt{\sigma}}\right|_{SU(3)}^{\text{pred}} \approx 6.15 \quad \text{vs. measured } 6.065(40),$$

i.e. agreement at 0.3σ (conservative model error) to $\sim 2\sigma$ (leave-one-out error); $SU(3)$ is the most-extrapolated, least-controlled point. With $\sqrt{\sigma} = 485(6)\,\text{MeV}$ this is $m_{1+-} \approx 2.98\,\text{GeV}$ predicted vs. $2.94\,\text{GeV}$ measured. The same fit yields falsifiable forward predictions at unmeasured ranks, $m_{1+-}/\sqrt{\sigma} \approx 5.77$ for $N = 14$–24: the ratio has effectively converged to its large-N limit by $N \simeq 14$, and a future lattice run there would test both the convergence and the $1/N^2$ form.

We stress the logical status: this is a held-out extrapolation of *lattice data* across gauge rank, made trustworthy by the exact-theory $1/N^2$ functional form. It is not a first-principles continuum mass from the y-series, which remains blocked as above.

11 Outlook

The fourth-order flatness and projected real-space geometry questions are now closed. Theorem 7.4 identifies the mobility operator as a local sum of squares of axial second differences and planar checkerboard differences. A remaining geometric refinement is to decompose the *unprojected* plaquette-space residual into boundary, co-boundary, and orthogonal sectors, clarifying which site-mediated processes mix the closed-surface subspace before projection.

The gauge-group dependence of the fourth-order band shape is closed for every rank: $N \geq 3$ by the band-shape theorem, and $N = 2$ by the exclusion Theorem 7.14. The determinant offsets q_4, q_6 are given, the $SU(3)$ band is carried to fifth order (Theorem 8.1), and the scale-matched mass/string-tension ratio is available through $O(y^5)$.

The open frontier is now sharply defined. First, the sixth-order rest mass m_6: its local algebra is cleared, but the connected six-insertion geometry census and global contraction remain an external-memory computation. Second, a project-native rerun of the fifth- and sixth-order string tensions σ_5, σ_6 (currently exact historical KPS targets). Third, the shell-six channel mixing needed before the isolated one-plaquette branch is identified with the physical lowest 1^{+-} state. None of these can alter the proven $N \geq 3$ band-shape results, and a continuum mass from the series itself (§10) remains blocked pending all three.

Finally, the interaction of two or more compact closed-surface excitations is open. Flat-band systems are often dominated by interaction effects; here the relevant question is whether the lower-order incidence protection produces bound states, constrained kinetics, or new collective sectors before the single-particle $O(y^4)$ mobility becomes important.

A Verification chain

Higher-order additions. The fifth-order coefficients q_5, A_5, B_5 (Theorem 8.1), the gate $c_R = 2c_M - c_X$, the rest-mass series (36), the ratio coefficients (37), and the $SU(2)$ identity $U^* = \varepsilon U \varepsilon^{-1}$ (Theorem 7.14) were each reproduced from source in exact arithmetic. The native string tension is exact through fifth order: σ_2, σ_3 in exact rational arithmetic and σ_4, σ_5 in exact finite-field arithmetic, each as a residue modulo three independent primes (Appendix B); σ_6 remains an exact KPS target and m_6 is open. The $1/N^2$ mass extrapolation (§10) uses external lattice data and is reported as such.

Every lower-order constant and structural claim retained from draft v0.1 is certified by the exact-arithmetic chain described there: `su3_moments_ext.py`, `su3_domino_d3.py`, `glueball_band_certificate_v2.py`, the two independent tromino checks, `cls_flat_band_certificate_v1_1.py`, and the document-level regression. That chain reported 387 hard gates and was rerun cold on June 13, 2026.

The complete from-scratch $SU(3)$ source chain has now been recovered and frozen from Stage 0 through Stages 1, 2, 3B, 3C, 3E, 3G, the folded des-Cloizeaux Stage 3I, and the final real-space assembly Stage 3J. Stage 3I is no longer a provenance gap. Its recovered Python source has SHA-256

`7662066c5516533ba531c414fc057c0f5c6e0d8bd1fe860a9c083fa8b2907abf`.

The folded-word payload consumed by Stage 3J has SHA-256

`973bf68f449c68f67cfc109928870175aa76d569a4e0b5fd787e35121b52eeb2`,

and this payload hash is stored in the 189-record kernel metadata and gated by the real-space SOS certificate.

The fourth-order analytic update is certified independently by `y4_exact_analytic_factorization_certificate.py`. It reads the complete 189-record real-space kernel in the same cell/origin gauge as (22) and adds 26 exact gates. These verify:

1. the kernel record count, orientation ordering, and real-space Hermiticity;
2. $H_4(0) = qI_3$ and the exact value (21);
3. the 25-term Laurent support, inversion symmetry, and reduction to the $3 + 3 + 6$ cubic frequency orbits;
4. cancellation of the term linear in X_i and the exact coefficients

$$A = \frac{5}{12}, \qquad B = \frac{17607806155349}{275331901291200};$$

5. the exact Γ, X, M, and R anchors;
6. the global-minimum and global-maximum hypotheses, bandwidth, and both band-edge curvature formulas.

The certificate status is `PASS`.

The real-space operator theorem is independently certified by `y4_exact_real_space_sos_certificate.py`. In exact Fraction arithmetic it forms $C^\dagger(H_4 - qI)C$ directly from all 189 kernel records, derives its 25 Laurent coefficients, and proves equality with (26). Its gates include exact Hermiticity, the scalar $H_4(0)$ value, the four stencil weights in (29), zero row sum, a finite-torus convolution regression, the $X/M/R$ generalized eigenvalues, and the exact bandwidth. The canonical semantic SHA-256 of the 189 sorted kernel records is

`48a422a517c7c1e70b84fd88a0773943f81ae3f9bfafadbe2304f8eb7d2e9b77`.

A separate single-block PyTorch script, `y4_sos_a100_validation.py`, applies both the full plaquette kernel and the 25-point scalar stencil on a periodic lattice and FFT-checks the numerator, Gram metric, generalized ratio, and parity-point lifts. This GPU run is an independent floating-point regression, not part of the exact proof.

The all-rank second-order extension is independently certified by `suN_closed_surface_band_stage1_certificate.py`, with 34 exact symbolic gates. It verifies the four channel weights, the closed form and positivity of t_N, the $SU(3)$ anchors, the incidence-kernel flat eigenvalue, and the high-symmetry reconstruction identities.

The stable-rank fourth-order extension is certified separately by `y4_sun_stable_rank_stage1.py`. Starting from the exact 182440-support Stage-0 geometry, it recomputes every sign mask after canonicalization and proves the counts in Table 1. It verifies the balanced Weingarten families through degree three, 37500 symbolic energy signatures, 94 distinct denominator polynomials, absence of accidental integer roots for $N \geq 7$, and an exact 140/140 regression against the $SU(3)$ balanced-channel engine. A cold rerun regenerated the Stage-0 support file with SHA-256

`5b3621cba76d4ac34b6ee89302399f07415db144735b07bbc926d0b370c4acdf`

and reproduced every stable-rank gate.

The local Stage-3C replacement is certified by `suN_walled_brauer_stage2a_exact.py` and the artifact `y4_sun_walled_brauer_local_projectors_exact.json.gz`. It constructs the 28 compact families and 134 path projectors of Theorem 7.10. The packaged matrices were independently re-parsed and all 746 Weingarten-entry gates, 1,338 projector-matrix gates, 730 denominator factors, and 140 history mappings were re-evaluated by `verify_wb_stage2a_serialized.py`. The deterministic projector archive has SHA-256

`a0e787e1e6619cab418ea1fedac209bbdc304c26e41bae9843902e7420debeb2`.

The global Stage-3G contraction and sign theorem are certified by `Y4_SUN_WALLED_BRAUER_FULL_SYMBOLIC_CERTIFICATE_2026-06-14.json` and `y4_sun_symbolic_qab_verify.py`. Exact partition/loop elimination reduces 3,850 trace topologies and 35,130 fusion paths to the three scalar outputs. The original product-denominator certificate proves

$$q_N = -\frac{2}{3N}\frac{Q_{32}(N^2)}{D_{34}(N^2)}, \qquad A_N = \frac{640}{N(N^2-1)^3}, \qquad B_N = \frac{P_{402}(N)}{D_{409}(N)},$$

with 33 and 403 positive Newton coefficients. The independent compact algebraic certificate then reduces the same 74-term B_N expression exactly to

$$B_N = \frac{P_{17}(N^2)}{NR_{20}(N^2)},$$

with 18 positive Newton coefficients about $z = 49$. The degree-409 expression remains a valid unreduced provenance certificate; the degree-17/20 form is the canonical rational theorem statement. Exact formulas match fixed-rank contractions at every $N = 7, \ldots, 18$, including an unused $N = 18$ holdout. Both verifiers pass unmodified. The full-symbolic certificate JSON has SHA-256

`50c3b0e945b87347416b1458b4346cee84b7d2e15385a152fa31a7adf1d500cd`.

The exceptional-rank theorem is certified by `y4_sun_all_n_ge3_band_shape_verify.py` and the associated $SU(4)$ cancellation ledger. The verifier checks the additional 312 $SU(4)$ sign assignments and 156 charge-conjugation orbits, exact $\Delta A_4 = \Delta B_4 = 0$, the full balanced $SU(4)$ regression, absence of $SU(5)$ determinant sectors, and exclusion of the sole $SU(6)$ determinant orbit from the A/B target. It ends with `ALL SU(N>=3) FOURTH-ORDER BAND-SHAPE GATES PASS`. This closes the band shape for every integer $N \geq 3$, while leaving only q_4, q_6 and the pseudoreal $SU(2)$ treatment open.

A.1 Canonical compact B_N polynomials

The reduced stable-rank and $N \geq 4$ band-shape formula is

$$B_N = \frac{P_{17}(z)}{NR_{20}(z)}, \qquad z = N^2,$$

where

$$\begin{aligned} P_{17}(z) = {} & 2096187310080z^{17} - 45206560309248z^{16} \\ & + 448972002607104z^{15} - 2723575470882816z^{14} \\ & + 11288692151812096z^{13} - 33888218411529728z^{12} \\ & + 76218901019673664z^{11} - 131068691814847264z^{10} \\ & + 174326341061538992z^{9} - 180230597250871976z^{8} \\ & + 144751635142984472z^{7} - 89742150515602808z^{6} \\ & + 42388925672412712z^{5} - 14916377727371552z^{4} \\ & + 3768794520714128z^{3} - 641987460459360z^{2} \\ & + 65414604672000z - 2967321600000, \end{aligned}$$

and

$$\begin{aligned} R_{20}(z) = {} & (z-1)^3(2z-3)(2z-1)^3(3z-2)(3z-1)(4z-9)^3(4z-5)(4z-1) \\ & \times (9z-25)(9z-16)(16z^2-44z+25)(16z^2-33z+16). \end{aligned}$$

The exact reduction cancels $(4N-5)(4N+5)(4N^2-3)$ from the unreduced product-denominator form.

The normative identity of the bundled fourth-order $SU(3)$ kernel is the canonical semantic hash of its 189 sorted records given above. Serialization- specific legacy hashes are not used as kernel identifiers.

The earlier interval subdivision, which resolved 3248 boxes with no unresolved leaves, remains an independent numerical cross-check but is not needed for the analytic global theorem.

B Native determinant-sector string tension and σ_5

The project-native torelon engine that fixes $\sigma_0, \ldots, \sigma_4$ contracts only the *adjoint* (balanced, $U(N)$ walled-Brauer) sector and therefore returns zero for every odd order: the odd-order and determinant contributions live entirely in the $SU(3)$ *determinant/ε* (triality) sector, in which three fundamental indices close through the Levi–Civita tensor ε_{ijk}. We have built an independent, from-scratch exact engine that treats both sectors uniformly. Each link's Haar integral is the orthogonal projector onto the $SU(3)$ singlet subspace of its index tensor $V^{\otimes a} \otimes \bar{V}^{\otimes b}$ (since $\int dU\, \rho(U)$ projects onto invariants), resolved into a fusion-tree basis by the nested cumulative quadratic Casimirs at the perturbative cuts; this yields simultaneously the color amplitude (a rational orthogonal projector) and the intermediate energies (Casimir eigenvalues) that feed the certified order-generic des-Cloizeaux folded weights. Because the $SU(3)$ Casimir assigns $C_2 = 0$ to the ε-singlet (a column of three boxes), the determinant sector is captured without special-casing. The single-link primitives are validated exactly: $\int dU\, U \otimes U^\dagger = \frac{1}{3}\delta\delta$ and $\int dU\, U \otimes U \otimes U = \frac{1}{6}\varepsilon\varepsilon$.

The engine reproduces every known order (reduced unit-vertex normalization, $\sigma_n^{\text{phys}} = (-1/4)^n \sigma_n^{\text{reduced}}$):

order	known (reduced)	this engine	status
σ_2	$-22/153$	$-22/153$	exact
σ_3	$61/408$	$61/408$	exact (first native
σ_4	$-737327120374220449/7250590288602460800$	exact mod 3 primes	**exact**
σ_5	$137767222189182735950309/2009803206414863779920000$	reconstructed, 7 primes	**exact** (independe

Here σ_2, σ_3 are obtained in *exact rational arithmetic* (the determinant sector, previously inaccessible, is closed for these) and are gate-grade outright. For σ_4 and σ_5 we certify exactness with an independent finite-field (GF(p)) realization of the same fusion-tree construction, reproducing the known value as a residue modulo three independent primes. The obstruction at σ_5 was the determinant links of local dimension $3^7 = 2187$: a direct GF(p) singlet projection at that dimension exceeds the in-core budget. It is removed by *weight-blocking*. The quadratic Casimir commutes with the $SU(3)$ Cartan, so the singlet subspace lies entirely in the *weight-zero block*, of dimension only 240 for the $(5, 2)$ sector (versus 2187); building the Casimir nullspace, the nested-Casimir simultaneous diagonalization, and the fusion-path projectors inside this block in pure integer arithmetic modulo p makes the exact evaluation tractable. The native engine evaluates σ_5 across all 22,820 canonical fifth-order torelon topologies (no bad prime) modulo *seven* independent primes $p \in \{33554467, 100000007, 134217757, 192999973, 192999949, 192999941, 192999931\}$; CRT combination (modulus $M \approx 6.25 \times 10^{56}$, 189 bits) and rational reconstruction then recover σ_5 *with no reference to any literature value*, uniquely (numerator and denominator of 77 and 81 bits, both below the $\sqrt{M/2}$ bound of 94 bits). The result coincides with the historical KPS value, used here only as an a-posteriori check. For σ_4, three $\sim 2^{25}$ primes give a $\sim$75-bit residue-match gate. Thus $\sigma_2, \ldots, \sigma_5$ are all exact, and σ_5 is an independent first-principles determination. The sixth-order tension σ_6 (three distinct plaquettes, double-ε links of local dimension 3^8) and the sixth-order rest mass m_6 remain external-memory–scale computations and are not asserted here.

C Domino benchmark

For a minimal exact benchmark (two plaquettes sharing one link, seven links): through the order considered, the one-flux levels of (1) are, in y^2 units above $8/3 \mp y$,

$$C\text{-even: } \left\{\tfrac{1769}{3060},\ \tfrac{13}{20}\right\}, \qquad C\text{-odd: } \left\{\tfrac{31}{68},\ \tfrac{17}{36}\right\}, \tag{39}$$

i.e. $\mathrm{diag} \pm |t|$ with $(\mathrm{diag}, |t|) = (\frac{1879}{3060}, \frac{11}{306})$ and $(\frac{71}{153}, \frac{5}{612})$ respectively. The upper C-even level sits exactly at the within-plaquette tower value $\frac{13}{20}$: in that combination the neighbor leakage cancels identically.

References

[1] J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Phys. Rev. D **11** (1975) 395.

[2] T. Banks, S. Raby, L. Susskind, J. Kogut, D. R. T. Jones, P. N. Scharbach and D. K. Sinclair, *Strong-coupling calculations of the hadron spectrum of quantum chromodynamics*, Phys. Rev. D **15** (1977) 1111.

[3] J. Kogut, D. K. Sinclair and L. Susskind, *A quantitative approach to low-energy quantum chromodynamics*, Nucl. Phys. B **114** (1976) 199.

[4] M. Creutz, *Quarks, Gluons and Lattices*, Cambridge University Press (1983).

[5] C. J. Morningstar and M. Peardon, *The glueball spectrum from an anisotropic lattice study*, Phys. Rev. D **60** (1999) 034509.

[6] A. Athenodorou and M. Teper, *The glueball spectrum of $SU(3)$ gauge theory in $3+1$ dimensions*, JHEP **11** (2020) 172.

[7] B. Sutherland, *Localization of electronic wave functions due to local topology*, Phys. Rev. B **34** (1986) 5208.

[8] E. H. Lieb, *Two theorems on the Hubbard model*, Phys. Rev. Lett. **62** (1989) 1201.

[9] A. Mielke, *Ferromagnetic ground states for the Hubbard model on line graphs*, J. Phys. A **24** (1991) L73.

[10] H. Tasaki, *Ferromagnetism in the Hubbard models with degenerate single-electron ground states*, Phys. Rev. Lett. **69** (1992) 1608.

[11] D. L. Bergman, C. Wu and L. Balents, *Band touching from real-space topology in frustrated hopping models*, Phys. Rev. B **78** (2008) 125104.

[12] D. Leykam, A. Andreanov and S. Flach, *Artificial flat band systems: from lattice models to experiments*, Adv. Phys. X **3** (2018) 1473052.

[13] C. W. Bauer et al., *Quantum simulation for high-energy physics*, PRX Quantum **4** (2023) 027001.

[14] A. Smith, *Two-sided spectral control of Wilson one-plaquette class gaps and an exactly verified strong-coupling bridge to the $SU(3)$ glueball channels*, companion manuscript (2026), with the Section 6 correction patch of June 11, 2026.